RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

NBA Accredited (UG - 6 Years
PG - 3 Years)

Department of Electronics & Communication Engineering

Date : 16/06/2026	IMPROVEMENT - CIE	Maximum Marks : 60
Semester : 2	UG	Duration : 2 Hrs
Course :	Principles of Electronics Engineering	Course Code :EC123ATC

SL. NO	Part-A Quiz	M	BT	CO
1.	Write the hexadecimal equivalent of the binary number 11101011?	1	1	1
2.	The minimum number of NAND gates required to realize XOR gates is ________	1	2	1
3.	Write the canonical sum-of-products (SOP) form for the Boolean function F(A, B) = A + B	1	3	2
4.	Determine the output of F = (A + B)(A' + C) when A=1, B=0, C=1.	1	3	2
5.	Realize a NOT gate using a NAND gate.	1	2	2
6.	Which component of an Embedded Linux system is responsible for initializing the hardware and loading the kernel into memory at startup?	1	2	2
7.	Small-scale embedded systems commonly use __________-bit microcontrollers.	1	1	2
8.	An engineer is building a prototype weather station and needs to perform advanced data analysis using MATLAB. Which IoT cloud platform is most suitable for this purpose?	1	2	1
9.	Give an example where Real time performance is a constraint for designing the embedded systems	1	1	1
10.	Write the 2's complement of $(11001)_2$	1	2	1

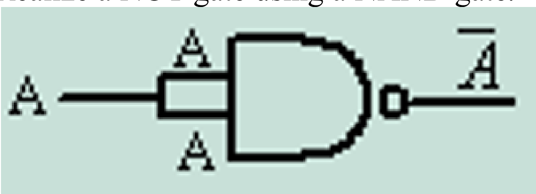
SL. NO	Part-B Test	M	BT	CO
1.a	Simplify the Boolean expressions and realize the simplified expression using basic gates. F1 = A'BC + ABC + AB'C F2 = A + A'B + AB	6	3	2
1.b	A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by one of the switches irrespective of the state of the other switch. State the logic of switching of the bulbs and explain.	4	4	3
2.a	Identify and discuss the key components of Embedded Linux.	6	2	1
2.b	Describe the Embedded system classification in terms of performance and function?	4	3	2
3.a	Illustrate how Edge-ML can be used in a wearable health monitor to process sensor data locally and generate real-time alerts without continuous cloud connectivity.	6	3	4
3.b	Explain the key design metrics with its key considerations to be considered during the design of embedded systems	4	3	3
4.a	Construct the truth table for a full adder and write expression for the same. Design a full adder using two half adders	6	3	4
4.b	Explain the hybrid architecture of Edge AI/ML systems, detailing the function of the device, edge, and cloud layers.	4	2	2

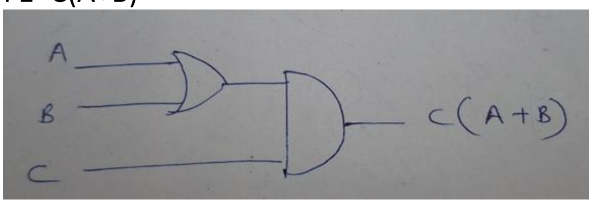
5.a	Simplify using K-map and implement using NAND gates: F(A,B,C,D) = $\Sigma m(1,3,4,6,9,11,12,14)$	6	3	2
5.b	Realize XNOR Gate using only minimum NOR Gates	4	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test-3	Max Marks-50	11	19	8	12	3	19	34	4	-	-

Date : 16/06/2026	IMPROVEMENT - CIE	Maximum Marks : 60
Semester : 2	UG	Duration : 2 Hrs
Course :	Principles of Electronics Engineering	Course Code: EC123ATC

SL. NO	Part-A Quiz	M	BT	CO
1.	Write the hexadecimal equivalent of the binary number 11101011? **EB**	1	1	1
2.	The minimum number of NAND gates required to realize XOR gates is ________ Ans.**4**	1	2	1
3.	Write the canonical sum-of-products (SOP) form for the Boolean function F(A, B) = A + B **A'B + AB' + AB (or Σm(1, 2, 3))**	1	3	2
4.	Determine the output of F = (A + B)(A' + C) when A=1, B=0, C=1. Ans: **1**	1	3	2
5.	Realize a NOT gate using a NAND gate. 	1	2	2
6.	Which component of an Embedded Linux system is responsible for initializing the hardware and loading the kernel into memory at startup? **BootLoader**	1	2	2
7.	Small-scale embedded systems commonly use __________-bit microcontrollers. **Ans: 8 or 16 bit**	1	1	2
8.	An engineer is building a prototype weather station and needs to perform advanced data analysis using MATLAB. Which IoT cloud platform is most suitable for this purpose? **ThingSpeak**	1	2	1
9.	Give an example where Real time performance is a constraint for designing the embedded systems **Ans: Automotive ECUs or network routers, any other example**	1	1	1
10.	Write the 2's complement of $(11001)_2$ **Ans:** $\mathbf{(00111)_2}$	1	2	1

Sl. No.	Solution with Mark split-up	Max marks
1.a	F1=C(A+B)  F2=A+B	6

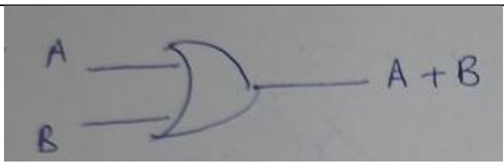


Diagram: 3M; Expression: 3M

1.b (4)

Truth table of EX-OR gate for two inputs

A B F
0 0 0
0 1 1
1 0 1
1 1 0 **2M**

The bulb can be put ON and OFF by any one of the switches.
Say,A is a ground floor switch, and if, A = 1, then it is ON, if A = 0, then it is OFF.
Say, Bis a first-floor switch, and if, B = 1, then it is ON, if, B = 0, then it is OFF.
F is the bulb and if F =1 is it is ON and if F=0, then it is OFF. Let both the switches are is off position (first row of truth table) the bulb is off (since F = 0). If switch (A) is put on (A = 1) then bulb turn on (F = 1) OR of switch (B) is put on then also bulb turns. Similarly, can verify for bulb to be off from either floor. **2M**

2.a (6)

Component	**Technical Description**	**Examples**
Bootloader	Initializes hardware, loads the kernel into memory during system startup.	U-Boot, Barebox
Linux Kernel	Core of the OS, managing processes, memory, I/O, and drivers; often customized and cross-compiled for target hardware.	Linux Kernel (mainline or vendor-specific)
Root Filesystem (RootFS)	Contains user-space utilities, libraries, and applications.	BusyBox, Buildroot, Yocto-based rootfs
Device Drivers	Interfaces between hardware peripherals and the OS for communication and control.	GPIO, UART, SPI, I2C drivers
Libraries & Middleware	Provides APIs for application development.	glibc, uClibc, musl
Application Layer	User-developed applications specific to the embedded product.	Industrial control apps, IoT data processing apps

6 components: 6 M

2.b (4)

Embedded systems can be classified based on Performance and Functional Requirements
1. Standalone Systems: Operate independently, without a network connection. Ex: Digital watches, MP3 players
2. Real-Time Systems: Deliver outputs within strict time constraints. • Hard Real-Time – Missing deadlines leads to system failure. • Soft Real-Time – Occasional deadline misses tolerable. Airbag controllers (Hard RT), Multimedia streaming (Soft RT)
3. Networked Systems: Connected via wired or wireless networks; support remote monitoring/control. Ex: IoT sensors, industrial PLCs
4. Mobile Embedded Systems: Designed for portable applications with mobility, often with energy constraints. Ex: Smartphones, wearables

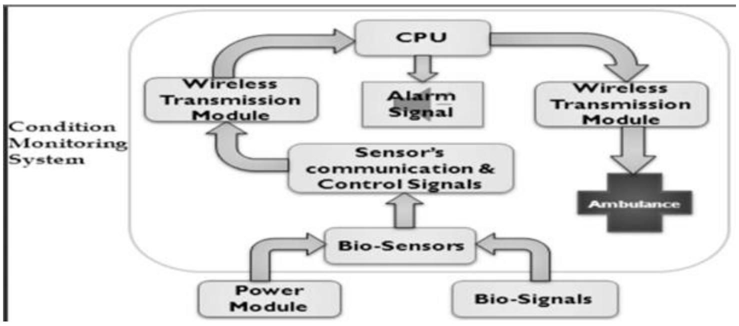
Explanation: 4M

3.a Wearable Health Monitors (Edge-AI Enabled) — **6**

Problem: Continuous, reliable tracking of physiological and activity signals with tight power, comfort, and privacy constraints.

Objectives:

1. Acquire PPG (heart rate/SpO_2), ECG (optional), accelerometer/gyroscope (activity/fall), skin temperature, GSR/EDA (stress).
2. Run on-device (edge) inference for HR, HRV, SpO_2, step count, activity class, and fall detection.
3. Guarantee 24–72 h battery life, BLE sync, and GDPR/HIPAA-ready data handling.



Block Diagram: 2M; Explanation: 4M

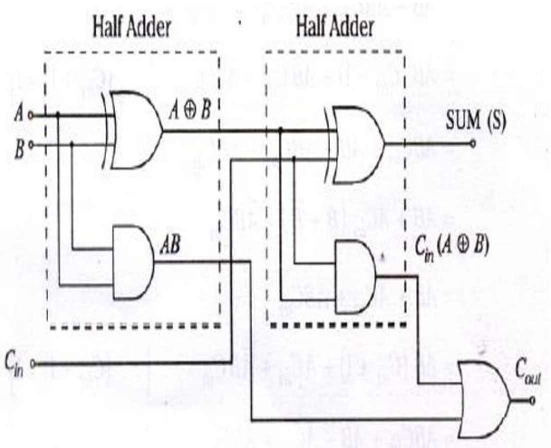
3.b — **4**

Design metrics are the quantitative parameters that guide the design, optimization, and evaluation of embedded systems.

Metric	Technical Description	Key Considerations
1. Power	Measures energy consumption in active, idle, and sleep modes. Vital for battery-powered and IoT devices.	Power budgeting, dynamic power management (DPM), use of low-power ICs, efficient clock gating.
2. Size	Physical footprint of the hardware (PCB area, SoC package size). Smaller size improves portability and integration.	PCB layout, system-on-chip (SoC) integration, MEMS, and packaging technologies.
3. Cost	Total system cost including hardware, software, and production. Critical in consumer electronics and large-volume markets.	Trade-off between performance and affordability; economies of scale; use of off-the-shelf components.
4. Performance	Ability to meet throughput, latency, and real-time deadlines efficiently.	Choice of microcontroller architecture, optimized code, pipelining, hardware accelerators, cache tuning.

4.a — **6**

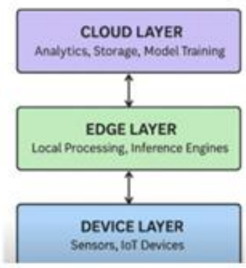
Input			Output	
A	B	C_{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Truth Table: 2M; Diagram: 2M; Expression: 1M

4.b

4



The architecture of Edge AI/ML systems typically follows a three-layer hybrid structure designed to balance low latency, scalability, and computational efficiency. This architecture integrates local processing capabilities with centralized cloud resources, enabling intelligent decision making near the data source.

The three layers are:

1**. Device Layer (Sensors and Data Generation)**

The Device Layer sits at the bottom of the architecture and includes sensors and IoT devices that are responsible for generating raw data.

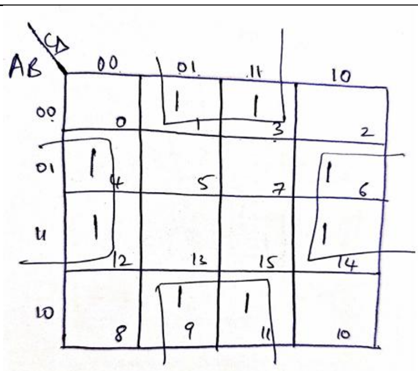
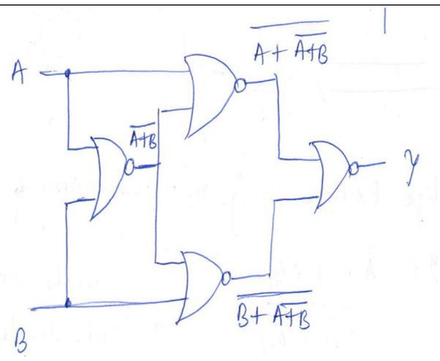
• Function: This layer focuses on data acquisition, capturing raw input such as images, telemetry signals, audio streams, or physiological measurements. These devices, often constrained by power and size, are the source of all information used by the system.

• Examples of Components: IoT sensors, embedded devices, or low-power microcontrollers (MCUs) like the ESP32.

2. **Edge Layer (Local Processing and Inference)**

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

NBA Accredited (UG - 6 Years
PG - 3 Years)

	The Edge Layer is the middle layer where the concept of Edge AI/ML is realized. Computation and storage occur physically close to the data source. • Function: This layer performs local processing and runs inference engines. It is responsible for preprocessing raw sensor data, executing optimized AI models in real-time to make predictions (inference), and filtering data. By processing data locally, it achieves ultra-fast response times and reduces network load. • Components: Edge nodes, gateways, or embedded boards (like Raspberry Pi or Arduino). These often utilize hardware accelerators (e.g., Google Coral TPU, NVIDIA Jetson) to provide high-speed computation for demanding inference tasks. 3. **Cloud Layer (Deep Analytics and Management)** The Cloud Layer forms the top of the architecture, providing centralized, high-performance computing resources and long-term data management. • Function: The cloud handles deeper analytics, long-term storage, and computationally heavy tasks. It is the typical location where datasets are large, and computational resources are abundant, making it suitable for model training and system updates. Processed summaries or logs may be sent back to the cloud for further analysis or performance tuning. • Components: Cloud backends like AWS IoT, Firebase, or ThingsBoard. Diagram: 2M; Explanation: 3M	
5.a	 $y = B\bar{D} + \bar{B}D$ $= B \oplus D$. K Map: 4M; Expression: 2M	6
5.b	 Diagram 2M and Proof 2M	4

Date : 16/06/2026	CIE- 3	Maximum Marks : 60
Semester : 2	UG	Duration : 2 Hrs
Course :	Principles of Electronics Engineering	Course Code : EC123ATC

SCRUTINY & EVALUATION OF CIE QUESTION PAPER

To be filled by the Course handling faculties:

As the faculty members of the course **Principles of Electronics Engineering** with code: **EC123ATC** we hereby confirm that the question paper is thoroughly reviewed and we ensured that it adheres to the following criteria:

S. NO	Criteria	Yes/ NO
1	The question paper adequately covers the prescribed syllabus contents.	Yes
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	Yes
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analyzing, evaluating, and creating.	Yes
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	Yes
5	Course handling faculty (As applicable) are responsible for preparing the question paper, scheme, and solution have unanimously agreed to utilize this question paper for conducting the Continuous Internal Evaluation.	Yes
6	The question paper, Scheme and Solution has been submitted to the Test coordinators within the designated timeframe to ensure the smooth conduction of Continuous Internal	Yes

Course handling Faculties

1. Dr. Shylashree N
2. Dr.Geetha K S and Prof. Sindhu Rajendran
3. Dr.Rajasree P M

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

NBA Accredited (UG - 6 Years
PG - 3 Years)

Department of Electronics & Communication Engineering

To be filled by the Scrutinizer's:

S. NO	Rubrics	Max points	Awarded
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams	10	
5	Scheme & Solution in the format	10	
	Total points	**50**	
Any other comments by the scrutinizers :			

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solution.

All the suggested corrections are incorporated and timely submitted	Yes/ NO	Accepted/ Rejected

Signature of Scrutinizer

(Name:)

Signature of HOD